ICARB-D ANNUAL SURVEY OF DENTAL SCHOOLS

A benchmark of health informatics in dental schools

Health informatics in dentistry: AI · Robotics · Bioinformatics · Software development

A guide for teachers and decision makers in dentistry



Questionnaire version 3.0.0

The main goal: a benchmark

Build a benchmark: one comparable number, from 0 to 100, that measures how mature each US dental school is in health informatics, with a score for each of its areas.

What the survey is for

1 MAIN GOAL

Where does each US dental school stand in health informatics? Every school gets a number built from its departments and courses, with a score for each area. The same questions also work as a checklist of what each course or department is missing to reach the maturity it wants: a guide for decision makers.

2 PRACTICAL CONCLUSION

If a grant for health informatics in dentistry were available, which university would have the best chance of using it well?

3 SECONDARY

How do schools compare with each other, and, where schools in other countries answer the same survey, how do US schools compare with them?

Health informatics is the umbrella

The benchmark measures health informatics as a whole. AI, robotics, bioinformatics and software development are areas inside it, next to its core: the information systems of the clinic.

Health informatics in dentistry · benchmark score 0–100

IS

Health information systems

Electronic records, imaging archives, coding, teledentistry and analytics

AI

Artificial intelligence

AI tools, AI teaching and generative AI

RO

Robotics and digital dentistry

Robots, simulators, scanners, CAD/CAM and 3D printing

BI

Bioinformatics

Omics laboratories, pipelines and biological data

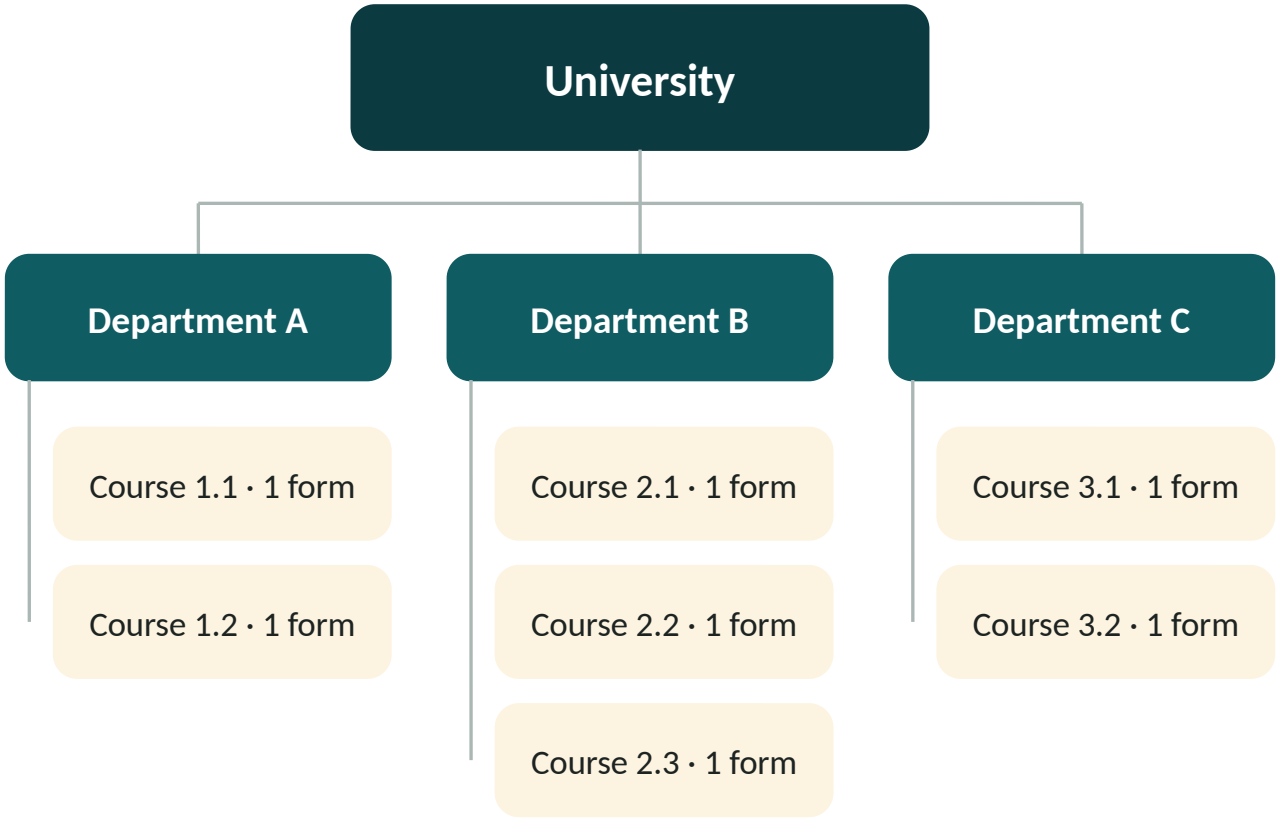
SW

Software development for health

Building, testing and maintaining health software

Foundations Infrastructure, data and privacy, people, governance, and implementation and sustainability. Every area depends on them.

Who answers: the head of each course



Course director answers

- Each course director answers one form for their course: nobody knows it better.
- Department score = mean of its courses.
- University benchmark = its departments, weighted by the number of courses.
- Many respondents per school make the benchmark more robust, and show where inside a school the strengths are.
- For shared resources (computers, clinic records, policies), directors answer for what their course can use; "Don't know" is allowed.

Why map maturity now

1

AI has reached the clinic

AI reads radiographs, digital workflows replace analogue ones, and students use generative AI every day.

2

Schools move at different speeds

Investment, training and rules vary widely, and no shared measure says who is ready or where the gaps are.

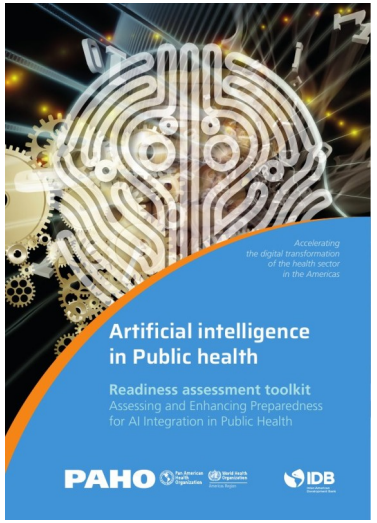
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Evidence to plan

A common, yearly measure lets deans and faculty plan, learn from the best prepared, and see progress.

Inspired by four PAHO publications

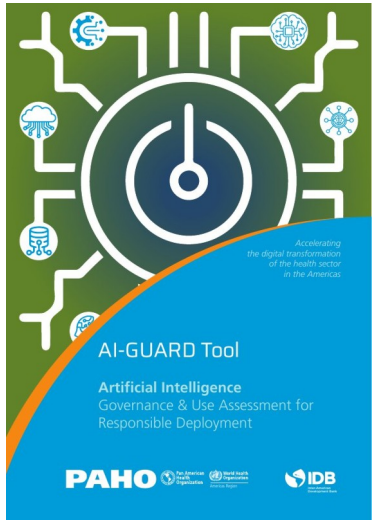
The Pan American Health Organization built a questionnaire to measure how ready public health institutions are to use AI. This survey adapts it to dental schools. Click a cover to open it on PAHO's website.



AI in Public Health: Readiness Toolkit

PAHO and IDB · 2024

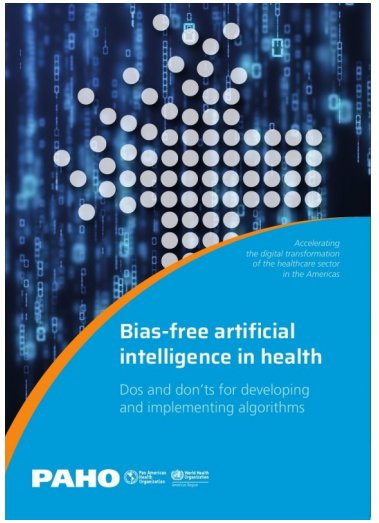
Main inspiration: the dimensions, the maturity scale and the analysis steps



AI-GUARD Tool

PAHO and IDB · 2026

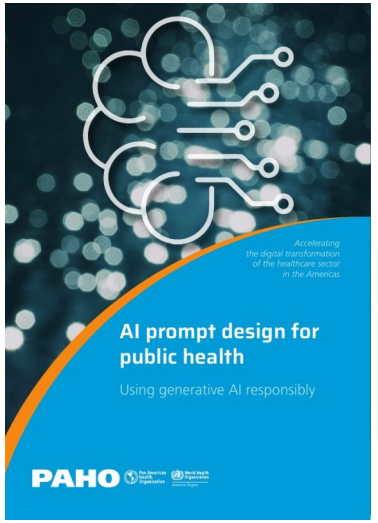
Governance checklist for AI tools (items F06 and J11)



Bias-free AI in Health

PAHO · 2026

Data representativeness and bias teaching (items C15 and J06)



AI Prompt Design for Public Health

PAHO · 2025

Prompt design and checking AI outputs (items F04–F05)

Thirteen groups of questions

180 questions in 13 parts. Part A is completed with the research team; parts B to M by the respondent, in about 45–60 minutes.

Who and what · Resources · Teaching · Areas of health informatics · Rules and lasting change · Self-view

A Profile of the course	B Context	C Physical resources	D Human resources
E AI curriculum	F Generative AI	G Information systems and software	H Robotics and digital dentistry
I Bioinformatics and omics	J Governance and ethics	K Implementation and sustainability	L Collaboration
M Self-assessment			

Part A · Profile of the course

A

16

questions

Completed with the research team

Who and what

GOAL

Place each course in its department and university, and record who answers for it, so every score can be traced, compared and followed from year to year.

What the questions cover

- Country, state and university
- Department and course (code and title)
- Training level and the respondent's role
- Survey year, consent and last year's code

Part B · Context

B

11
questions

Answered by the respondent

Who and what

GOAL

Measure the size of the course, so every count becomes a fair ratio (per student, per faculty member, per year).

What the questions cover

- Students and faculty (headcount and full-time equivalent)
- Academic calendar and terms per year
- Course hours and the areas of health informatics it touches

Part C · Physical resources

C

22
questions

Answered by the respondent

Resources

GOAL

Know what the course has today: computing, software, AI tools, data, privacy and security.

What the questions cover

- Computers, GPUs, cloud access and dedicated spaces
- Software and the AI tools students actually use
- Datasets, data quality and representativeness
- Privacy, security, connectivity and IT support

Part D · Human resources

D

16
questions

Answered by the respondent

Resources

GOAL

Know who can teach, build and support these technologies, and what holds them back.

What the questions cover

- Faculty able to teach AI, and faculty doing AI research
- Training received and planned
- Partners and technical staff
- Openness to change and the main barriers

Part E · AI curriculum

E

24

questions

Answered by the respondent

Teaching

GOAL

Know what students actually learn about AI, at which level and for how many hours.

What the questions cover

- Hours per year, by teaching format and by 16 topics
- Programming and formal AI model development
- Named predoctoral and graduate courses
- Assessment, materials and planned changes

Part F · Generative AI

F

8

questions

Answered by the respondent

Teaching

GOAL

Know whether generative AI is used with clear rules, real skills and working safeguards.

What the questions cover

- Policy and permitted uses for students and faculty
- Prompt design and checking AI outputs
- Safeguards in place (adapted from PAHO AI-GUARD)

Part G · Information systems and software

G

16
questions

Answered by the respondent

Areas of health informatics

GOAL

Know whether the clinic produces reliable digital data, and whether the course can build and maintain its own health software.

What the questions cover

- Electronic records, imaging archives, coding and teledentistry
- Access to clinical data, teaching hours and informatics staff
- Software development for health: what is built and how
- Teaching students to design and build health software

Part H · Robotics and digital dentistry

H

10
questions

Answered by the respondent

Areas of health informatics

GOAL

Know what robotic and digital equipment exists, and whether students really use it.

What the questions cover

- Number of robots, simulators, scanners, CBCT, milling units and 3D printers
- How students get hands-on access
- Digital workflow, maintenance and trained faculty
- Teaching hours and research

Part I · Bioinformatics and omics

I

10

questions

Answered by the respondent

Areas of health informatics

GOAL

Know whether the course can produce, analyse and share biological data.

What the questions cover

- Laboratories, sequencing, biobanks and computing
- Pipelines and people with bioinformatics skills
- Teaching hours and research areas
- How data are shared (FAIR principles)

Part J · Governance and ethics

J

12

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether technology is used safely, ethically and under clear responsibility.

What the questions cover

- Written plan, leadership and oversight
- Ethics and bias teaching
- Standards and guidance in use (ADA, CODA, HIPAA, FDA, NIST)
- Patient consent and a governance checklist for AI tools

Part K · Implementation and sustainability

K

14

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether projects move from pilot to routine use, are evaluated, and can last.

What the questions cover

- Pilot projects and whether they scale
- Metrics, evaluation and feedback
- Cost-benefit and sustainability
- Funding sources, budget and grants

Part L · Collaboration

L

8

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether the course is ready to share data and work with other institutions on AI.

What the questions cover

- Interest in multi-institution AI projects
- De-identified data that could be shared
- Preferred ways to collaborate
- What stands in the way

Part M · Self-assessment

M

13

questions

Answered by the respondent

Self-view

GOAL

Compare how the course sees itself with what its answers show, and record its most urgent needs.

What the questions cover

- Overall maturity in health informatics and each of its areas
- Main barriers and facilitators
- The single most urgent need
- Innovative practice and the vision for 2030

From items to one number

Items are rated from 1 to 4 ("Don't know" is never counted as zero). Each domain becomes a 0–100 index, and the domain indices combine into the benchmark: course → department → university.

1

Absent

Nothing in place yet.

2

Initial

Interest or informal, isolated efforts.

3

Developing

In place in part, or not yet routine.

4

Established

Routine, documented and supported.

Who could best use a grant?

If a grant for health informatics in dentistry were available, the results point to the universities most likely to use it well.

1

Benchmark

How mature the school already is in health informatics (0–100), with a score for each area.

+

2

Funding readiness

The strength of its foundations: infrastructure, data, people, governance, and turning pilots into practice.

=

★

Best placed to invest

Schools strong on both are the most likely to turn new money into lasting results.

Comparing schools, and comparing US schools with universities abroad that use the same survey, are secondary uses of the same numbers.

After the survey: a checklist for improvement

A guide for decision makers: the same items become 91 checkpoints that show what each course or department is missing. For each one, the target is level 4 and the next step is level 3.

1

Answer

Each course director completes the survey for their course.

2

See where you stand

Read the benchmark and the score for each area.

3

Choose a target

Decide the level you want to reach, area by area.

4

Tick the checklist

Mark what is already true; the rest are your next steps.

5

Measure again

Answer next year and see the progress.

The full checklist is online at igoralves1.github.io/AI-assessment-DS-2026/#/questions?go=checklist and in chapter 4 of the questionnaire (PDF).

One number, every year

Every question, what it captures and how it is scored:
igoralves1.github.io/AI-assessment-DS-2026/#/questions

